

INTRODUCTION PROJECT

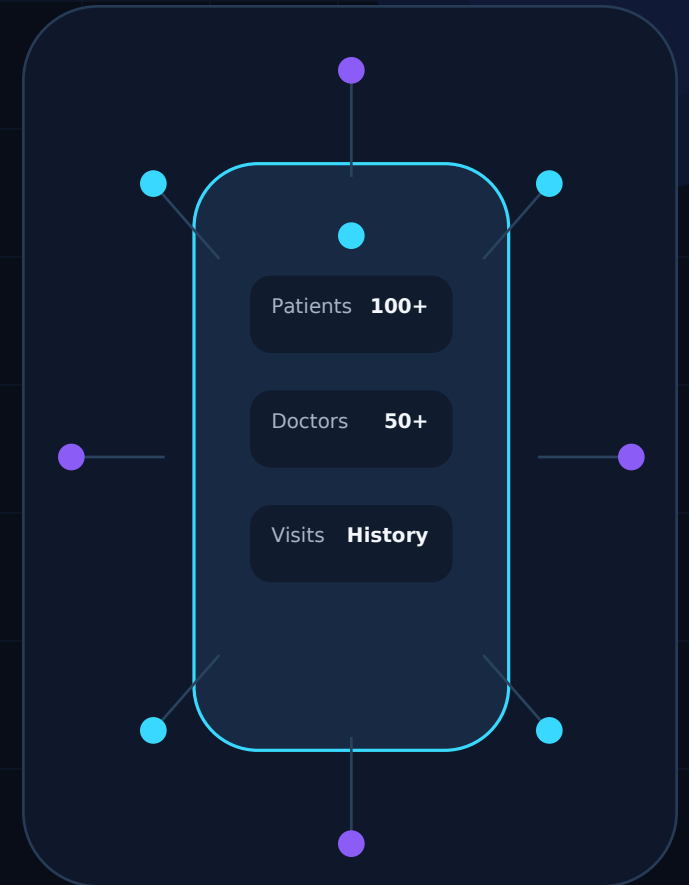
# Clinic Management System

A console-based C++ system for managing patients, doctors, waiting priorities, and visit history.

Team Members

**Kareem Hammad**

C++ • OOP • Data Structures • Algorithms



# What does the system do?

## ● CORE FEATURES

### Patients

Add, view, search by ID, delete, sort

### Doctors

Add and display doctor information

### Waiting Room

Emergency priority + arrival order

### Visits

Record diagnosis, doctor, date and fee

### History

Display visits and calculate total paid

## ● SYSTEM FLOW

01

### Register

Patient / Doctor

02

### Queue

Emergency / Normal

03

### Call Next

Priority-based

04

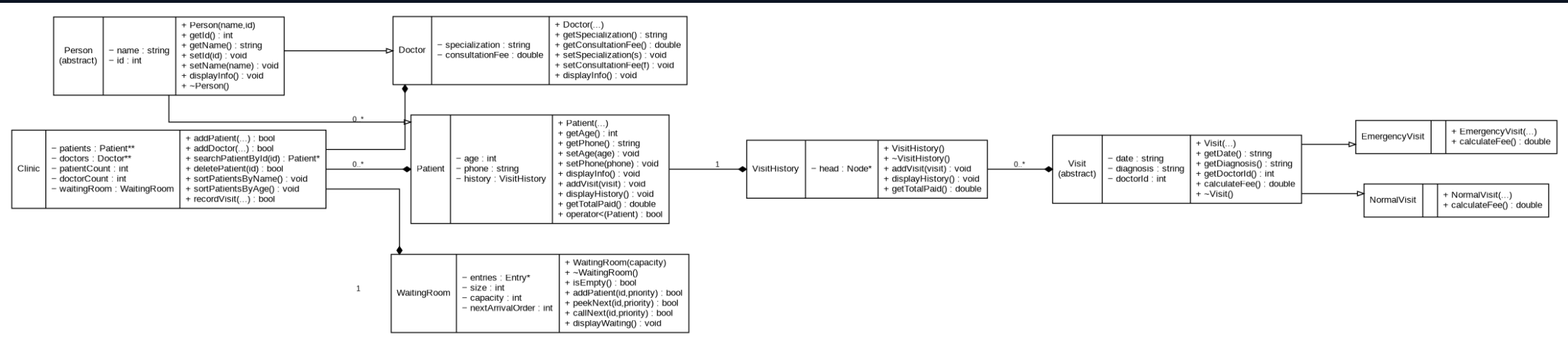
### Record Visit

Doctor + Diagnosis

05

### History

Fees + Total



The diagram shows the main classes, inheritance, and relationships used by the implementation.

# Object-Oriented Design

## ABSTRACTION

### Person

Abstract base class  
with pure virtual  
displayInfo()

## INHERITANCE

### Patient / Doctor

Both inherit common  
identity data from  
Person

## POLYMORPHISM

### Visit

EmergencyVisit and  
NormalVisit override  
calculateFee()

## ENCAPSULATION

### Private Data

Access through  
getters / setters

```
Visit* visit = new EmergencyVisit(...);  
double fee = visit->calculateFee(); // runtime dispatch
```

# Data Structures & Algorithms

## ● DATA STRUCTURES

- Custom linked list for VisitHistory  
Stores visit records dynamically.
- Custom priority queue for WaitingRoom  
Emergency first, then arrival order.
- Dynamic pointer arrays for patients/doctors  
Keeps patient IDs sorted for binary search.

## ● ALGORITHMS

- Binary Search Search patient by ID
- Selection Sort Sort by name / age
- Recursion Calculate total paid
- Comparison Count Measure sorting comparisons

All required structures are implemented from scratch instead of using `std::vector` / `std::queue` / `std::list` for the required DS.

# Complexity at a Glance

| Operation             | Big O       |
|-----------------------|-------------|
| Add Patient           | $O(n)$      |
| Search Patient        | $O(\log n)$ |
| Delete Patient        | $O(n)$      |
| Add to Waiting Room   | $O(n)$      |
| Peek Next Patient     | $O(1)$      |
| Call Next Patient     | $O(n)$      |
| Add Visit             | $O(1)$      |
| Visit History / Total | $O(v)$      |
| Sort Patients         | $O(n^2)$    |

## KEY IDEA

### Fast patient lookup

comes from keeping  
patients sorted by ID

**n** = patients    **v** = visits

# Demo & Delivery

## ● LIVE DEMO FLOW

- 1 Add patient + doctor
- 2 Add patients to waiting room
- 3 Show Emergency before Normal
- 4 Call next patient
- 5 Record visit and fee
- 6 Display history + total

## ● PROJECT INFO

### Repository

[\[GitHub repository link\]](#)

### Team

Kareem Hammad

### Built with

C++

OOP

DSA

# Thank you

Questions & live demonstration